
Curation Metrics Are a Post-Training Decision: Auditing Trajectory Smoothness for Adapting Robot Foundation Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Trajectory smoothness has been proposed as a cheap, training-free signal for curat-
2 ing imitation-learning demonstrations, most recently by RINSE, which introduces
3 Spectral Arc Length (SAL) and a contact-aware Trajectory-Envelope Distance
4 (TED) and reports downstream gains on robomimic. Recent audit work showed
5 that generic curation metrics on a synthetic-defect LIBERO testbed largely exploit
6 episode length rather than demonstration content. We test whether the *specific*
7 *published metrics* survive the corresponding controls on *real* data. On robomimic
8 can/mh, SAL separates better from worse operators with AUROC 0.952, but
9 episode length alone achieves 0.957, and inside the tightest duration-matched
10 band SAL falls to 0.628 while raw mean jerk, the baseline RINSE argues against,
11 rises to 0.896. On can/mg, where every episode is exactly 150 steps and length
12 cannot carry information, SAL is at chance (0.454) and TED is inverted (0.376)
13 against the task-success label, while jerk reaches 0.706. Training GMM behavior-
14 cloning policies on each metric’s curated subset (10 seeds per condition), curating
15 by TED costs roughly a third of downstream success relative to random selection
16 of the same size (0.240 vs. 0.360, $p=0.0001$, surviving Bonferroni correction);
17 path length is directionally harmful with weaker support, while SAL and jerk are
18 not statistically distinguishable from not curating at all. Replicating the grid under
19 a diffusion policy, TED remains the worst condition, but the two architectures do
20 not agree on the rest: jerk, inert under behavior cloning, becomes significantly
21 *better* than both baselines under diffusion, so which metric a pipeline should trust
22 is not a property of the metric alone. Inspecting individual trajectories shows
23 why: on this task successful manipulation requires sustained high-deviation motion
24 through the contact phase, while failure manifests as smooth low-amplitude
25 hovering, so a smoothness metric measures commitment with the sign flipped.
26 The length confound replicates on a second task: on square/mh, SAL correlates
27 $+0.96$ with episode duration and its skill AUROC falls from 0.898 to 0.480 in-
28 side a duration-matched band, while jerk rises from 0.431 to 0.896. For post-
29 training pipelines that filter, reweight, or select demonstrations before fine-tuning
30 a pretrained policy, the practical consequence is direct: the curation step can
31 silently discard the corrective, high-effort demonstrations that fine-tuning most
32 needs, while keeping the smooth, low-information ones. Our results extend the
33 detection-vs-curation decoupling to named, published metrics, on operator labels
34 rather than injected defects, and to a benchmark where the length confound is
35 excluded by construction rather than by truncation.

1 Introduction

A large share of post-training work for robot foundation models runs on curated demonstration subsets: filtering, reweighting, or ranking trajectories before fine-tuning, whether the source is human teleoperation, autonomous rollouts, or a mix of both. The curation step is rarely evaluated as carefully as the fine-tuning step that follows it, and the metrics used to perform it are typically validated once, on clean data, and then reused across deployments without re-checking whether they still measure what they claim to measure. This paper is a case study in what goes wrong when that assumption fails: a published, named curation metric that looks reliable on successful demonstrations, and is actively harmful once the pool it filters contains the failures and near-misses that post-training adaptation pipelines actually have to work with.

Behavior cloning inherits the quality of its demonstrations, and a growing literature proposes to score and filter demonstrations automatically. Training-free trajectory statistics are the cheapest family of such scores, and the most developed recent proposal is RINSE [Kulkarni et al., 2026], which ranks demonstrations by two smoothness metrics: Spectral Arc Length (SAL), a frequency-domain measure adapted from rehabilitation science, and Trajectory-Envelope Distance (TED), a contact-aware spatial deviation from a Bézier-smoothed reference. RINSE reports that filtering by these metrics matches or beats full-data training on robomimic tasks.

A recent line of audit work complicates this picture. Bedi [Bedi, 2026a] showed on a controlled testbed that action-only curation scorers are blind or inverted on structural defects, and the follow-up [Bedi, 2026b] showed that five of seven curation metrics on a LIBERO testbed achieved near-perfect detection AUROC by measuring episode length rather than demonstration content, and that detection accuracy and curation value are sharply decoupled. Those results, however, were obtained with *generic* metric implementations (a SPARC-style smoothness score among them), with *synthetically injected* defects, and with episode length controlled by *truncation*.

This paper asks the natural next question: do the specific, published metrics of RINSE survive these controls on real data? Our audit differs from the Bedi testbeds on three axes, each of which closes a gap that line of work explicitly left open:

1. **Named metrics, not proxies.** We implement SAL from its published formula and TED from the RINSE appendix, including the contact-boundary partition that is TED’s distinguishing design feature. TED has not previously been evaluated outside its authors’ own success-filtered setting.
2. **Real labels, not injected defects.** robomimic can/mh [Mandlekar et al., 2021] carries *human operator skill tiers* (better/okay/worse) assigned to real teleoperators, and can/mg carries real task-success outcomes of a scripted-policy data-generation process. No defect is synthetic.
3. **A length-free benchmark by construction.** In can/mg every episode is exactly 150 steps. The episode-length confound documented by [Bedi, 2026b] cannot exist in this data, with no truncation required. Truncation removes the confound by discarding signal; a fixed-horizon dataset removes it without touching the trajectories.

Our findings are easy to state. Smoothness metrics rank *operator skill* largely to the extent that skill is confounded with how long the episode took. Once duration is controlled, by normalization or by duration-matched subsets, SAL’s advantage collapses and raw mean jerk, the crude baseline the smoothness literature positions itself against, becomes the strongest detector. Against *task success*, on fixed-length data, both of RINSE’s metrics are at or below chance, with TED inverted, while jerk retains signal. Curating by TED produces a policy substantially worse than random subsampling of the same size, a result that survives correction for multiple comparisons; the weaker metrics are not distinguishable from doing nothing. Skill and success are different targets, and the curation literature validates on one while deploying against the other.

2 Related Work

Trajectory-statistic curation. RINSE [Kulkarni et al., 2026] is the primary subject of this audit; it assumes demonstration pools pre-filtered for task success and validates on successful trajectories

only, a scoping decision its limitations section states explicitly. PSD [Sojib and Begum, 2026] scores demonstrations by total spectral power of the end-effector trajectory and reports gains over learning-based curation on robomimic-MH and a non-expert dataset. Neither work evaluates on pools containing failures.

Audits of curation metrics. Bedi’s testbed papers [Bedi, 2026a,b,c] established the detection-vs-curation decoupling, documented the episode-length confound, and reported a negative result on per-phase metric selection. We adopt their framing and extend it to published metrics, real labels, and a natively fixed-length benchmark. Where they control length by truncating to a fixed prefix, our can/mg experiments require no such intervention.

Evaluation practice. Kress-Gazit et al. [Kress-Gazit et al., 2024] and RoboEval [Wang et al., 2026] argue that binary success rate under-describes policy quality, and [Jiang et al., 2026] shows that widely used benchmarks admit shortcut solutions. Our results are a data-side complement: the scores used to *select training data* can also be shortcut-driven, with episode duration as the shortcut.

3 Metrics and Datasets

Metrics. All metrics are computed from proprioceptive signals at the robomimic control rate of 20 Hz ($\Delta t = 0.05$ s) and return a badness score (higher is worse). *SAL*: spectral arc length of the end-effector speed profile, implemented from the published formula; we negate so that higher means less smooth. *TED*: contact-aware trajectory-envelope distance, reimplemented from the RINSE appendix: the 6-D pose trajectory (position plus rotation vector, orientation weight $w_{\text{ori}} = 0.25$) is partitioned at gripper open/close transitions, each segment is fit with a greedy Bézier envelope, and the score is the length-normalized DTW distance between trajectory and envelope. *Jerk*: mean third-difference magnitude of end-effector position. *Path length*: total end-effector arc length. Where indicated we also report per-timestep normalizations (score divided by $T-1$). Our TED is a reimplement from the paper, not the authors’ code; Section 7 discusses the resulting caveat and the sanity check we ran.

Datasets. All data is public robomimic [Mandlekar et al., 2021] low-dim. *can/mh*: 300 successful demonstrations from six human operators with released skill tiers (better/okay/worse, 100 each); episode lengths range 98–1050 steps. *can/mg*: 3900 machine-generated demonstrations, 18.4% successful, every episode exactly 150 steps. *can/paired*: 200 demonstrations in matched success/failure pairs (50% successful), lengths 55–147; we report it as a secondary setting because its matched-pair construction makes it adversarially hard for trajectory-level scores. *square/mh*: 300 successful demonstrations with the same skill-tier structure (100 each), episode lengths 123–1051, used in Section 4.4 to test whether the length confound is task-specific. We note that Can is the only robomimic task with failure-containing splits (mg and paired); Square, Transport, and Tool Hang ship only proficient-human and multi-human data, all of it successful. Our success-label experiments are therefore confined to Can by benchmark availability, not by choice.

Statistics. All AUROCs are reported with 95% bootstrap confidence intervals (2000 resamples). Downstream comparisons use difference-of-means bootstrap CIs (20,000 resamples) and Welch two-sample tests, with Bonferroni correction applied across the full set of reported comparisons; we mark which results survive correction and which do not. For skill we report better-vs-worse AUROC; for success we report AUROC against the binary success label with scores negated so that $\text{AUROC} > 0.5$ means the metric ranks clean/successful demonstrations higher, and $\text{AUROC} < 0.5$ means the metric is *inverted*.

4 Detection Results

4.1 Operator skill on can/mh: the length confound, on real labels

On the full 200 better/worse demonstrations, SAL appears excellent: AUROC 0.952 [0.922, 0.975]. But episode length alone achieves 0.957 [0.929, 0.980], and SAL’s correlation with T is +0.97. The tier means make the mechanism plain: worse operators average 304 steps, better operators 143, and SAL’s tier means (100.4 / 57.5 / 44.9) track duration almost linearly. On this dataset SAL is, to first order, a stopwatch.

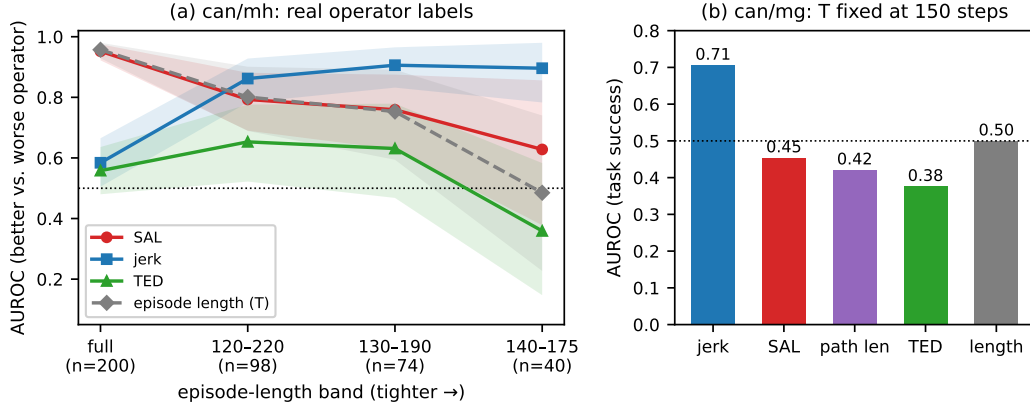


Figure 1: **Smoothness metrics rank duration, not quality.** (a) Better-vs-worse operator AUROC on can/mh across duration bands of increasing tightness (shaded: 95% bootstrap CI). SAL (red) tracks the episode-length baseline (gray, dashed) almost exactly and falls from 0.952 to 0.628 as the band tightens; raw jerk (blue) moves the opposite way, 0.584 to 0.896; TED (green) never clears chance and inverts in the tightest band. In that band episode length itself is at chance (0.485), confirming the control. (b) On can/mg every episode is exactly 150 steps, so length carries no information by construction; against the task-success label, jerk retains signal (0.706), SAL is at chance (0.454), and TED is inverted (0.376).

	full ($n=200$)	$T \in [120, 220]$ ($n=98$)	$T \in [130, 190]$ ($n=74$)	$T \in [140, 175]$ ($n=40$)
SAL	0.952 [.922,.975]	0.793 [.690,.881]	0.759 [.628,.875]	0.628 [.378,.856]
SAL/ T	0.582 [.497,.662]	0.600 [.479,.715]	0.603 [.446,.755]	0.658 [.342,.923]
TED	0.558 [.480,.636]	0.653 [.522,.776]	0.631 [.468,.779]	0.359 [.147,.583]
jerk	0.584 [.505,.665]	0.862 [.783,.928]	0.906 [.832,.965]	0.896 [.783,.980]
path len.	0.783 [.718,.842]	0.590 [.449,.730]	0.624 [.461,.781]	0.429 [.215,.645]
length T	0.957 [.929,.980]	0.801 [.689,.901]	0.753 [.594,.891]	0.485 [.227,.740]

Table 1: Better-vs-worse operator AUROC on can/mh across duration bands, with 95% bootstrap CIs. The bottom row is the control: a band is duration-matched only where T itself is near chance. Per-timestep normalizations of TED and path length are omitted: unlike SAL, whose raw value grows with the number of spectral bins, these quantities do not scale linearly with T , and dividing by T turns them into inverse-speed proxies rather than duration-free measures of smoothness.

Two independent controls confirm this. First, *normalization*: SAL’s arc length sums over $T/2$ spectral bins, so dividing by $T-1$ is the principled per-timestep correction; normalized SAL drops from 0.952 to 0.582 [0.497, 0.662] and its length correlation drops from +0.97 to +0.07. Second, *duration matching*: restricting to bands of increasingly similar episode length (Figure 1a, Table 1). In the tightest band ($T \in [140, 175]$, $n=40$), where length itself is at chance (0.485 [0.227, 0.740]), SAL is 0.628 [0.378, 0.856], a CI that includes chance, while raw jerk is 0.896 [0.783, 0.980]. Across the four bands the two metrics *cross*: as the confound is removed, the crude metric strengthens monotonically (0.584, 0.862, 0.906, 0.896) and the sophisticated one weakens monotonically (0.952, 0.793, 0.759, 0.628).

TED never clears chance in any band (0.558 full-sample, upper CI 0.636) and inverts to 0.359 in the tightest band. TED does *not* have the length pathology (correlation +0.14 with T); it simply carries little skill signal on this data.

We caution that the tightest band retains only 7 worse-operator demonstrations and its intervals are wide; the claim rests on the *trend* across all four bands and its agreement with the normalization control, not on any single cell. We also note that naïve per-timestep normalization is only appropriate for metrics whose raw value grows with T ; normalized path length, for example, becomes an inverse-speed proxy that *rises* with skill and is reported only for completeness in Table 1.

154 4.2 Task success on can/mg: the confound removed by construction

155 The can/mg split fixes every episode at 150 steps, so no metric can exploit duration. Against the
156 success label (716 of 3900 successful), jerk achieves 0.706, SAL 0.454, path length 0.421, and TED
157 0.376 (Figure 1b). The length baseline is exactly 0.500, as it must be. Per-timestep normaliza-
158 tion leaves every AUROC unchanged, as expected under a constant divisor, which we report as a
159 consistency check.

160 Two readings of this result matter. First, both of RINSE’s metrics are at or below chance on the
161 label practitioners actually deploy against. TED’s inversion means that filtering by TED *preferen-*
162 *tially retains failed demonstrations*. Second, the ordering among metrics reverses relative to the
163 uncontrolled skill setting: the metric that looked worst there (jerk) is the only one with signal here.

164 4.3 Secondary setting: can/paired

165 On the matched-pair dataset every metric is inverted against success (TED 0.331, SAL 0.398, jerk
166 0.368, path length 0.029), but the setting is confounded in the opposite direction: successful demon-
167 strations are systematically *longer* (length-alone AUROC 0.194), and path length’s $+0.86$ correla-
168 tion with T makes it a pure length detector here. We report can/paired for completeness and treat
169 can/mg as the primary success-label evidence.

170 4.4 Generalization to a second task: square/mh

171 If the length confound were an artifact of the Can task, the argument would be considerably weaker.
172 We therefore repeat the skill-label analysis on square/mh, a harder insertion task with the same
173 operator skill-tier structure and a comparable duration range (123–1051 steps versus Can’s 98–
174 1050). No retuning was performed: the same metric implementations and the same band procedure
175 were applied to the new data.

176 The confound replicates almost exactly. SAL’s correlation with episode duration is $+0.96$ on Square
177 against $+0.97$ on Can, and its uncontrolled skill AUROC is 0.898 [0.852, 0.936] against a length-
178 alone baseline of 0.936 [0.902, 0.964]. Per-timestep normalization collapses it to 0.458 [0.376,
179 0.536], slightly below chance and a larger drop than the corresponding Can figure. Path length
180 behaves the same way, reaching 0.931 [0.893, 0.963] uncontrolled with a $+0.90$ duration correlation.

181 The crossover also replicates (Table 2). As the duration band tightens and the length baseline
182 falls from 0.936 to 0.407, SAL falls monotonically (0.898, 0.759, 0.529, 0.480) while jerk rises
183 monotonically (0.431, 0.570, 0.782, 0.896). In the band where duration is genuinely controlled
184 ($T \in [200, 240]$, length AUROC 0.407), jerk reaches 0.896 [0.724, 1.000], numerically identical to
185 its value in Can’s tightest band, while SAL sits at 0.480 [0.250, 0.701], a CI centred on chance.

186 Two differences from Can are worth stating. First, jerk is *inverted* on Square before any control is
187 applied (0.431), where on Can it was weakly positive (0.584); the uncontrolled ranking of metrics
188 is therefore itself task-dependent, even though the controlled ranking is not. Second, path length
189 remains strong in the controlled band on Square (0.887 [0.742, 0.991]) where it collapsed on Can
190 (0.429 [0.215, 0.645]). We do not claim jerk or path length is a good curation metric on the strength
191 of these numbers; the point is narrower, that whatever advantage SAL displays over them disappears
192 once duration is held fixed, on both tasks. The tightest Square band retains 13 worse-operator
193 demonstrations, so as with Can the claim rests on the monotone trend across bands rather than on
194 any single cell.

195 5 Does Detection Predict Curation? Downstream Training

196 Detection AUROC is not the quantity practitioners care about [Bedi, 2026b]; the trained policy is.
197 We therefore filter can/mg by each metric at a 50% keep fraction and train a GMM behavior-cloning
198 policy (MLP 1024×1024 , 5 modes, batch 100, 600 epochs of 500 gradient steps, learning rate
199 10^{-4}) on each curated subset, with 50 evaluation rollouts (horizon 400) every 150 epochs and the
200 best checkpoint reported per seed. Conditions: TED, SAL, jerk, path length, plus three references:
201 an *oracle* keeping only successful demonstrations, a *random* 50% subset, and an uncuration size-
202 matched control. Ten seeds per condition. All filtered pools contain 1950 demonstrations except the

	full ($n=200$)	$T \in [150, 300]$ ($n=111$)	$T \in [180, 260]$ ($n=55$)	$T \in [200, 240]$ ($n=30$)
SAL	0.898 [.852,.936]	0.759 [.658,.849]	0.529 [.344,.702]	0.480 [.250,.701]
SAL/ T	0.458 [.376,.536]	0.485 [.372,.602]	0.468 [.300,.653]	0.520 [.280,.738]
TED	0.663 [.584,.736]	0.685 [.577,.788]	0.599 [.427,.758]	0.557 [.335,.785]
jerk	0.431 [.349,.509]	0.570 [.455,.687]	0.782 [.618,.923]	0.896 [.724,1.000]
path len.	0.931 [.893,.963]	0.817 [.726,.895]	0.796 [.654,.913]	0.887 [.742,.991]
length T	0.936 [.902,.964]	0.824 [.738,.898]	0.613 [.456,.766]	0.407 [.197,.629]

Table 2: Better-vs-worse operator AUROC on *square/mh* across duration bands, with 95% bootstrap CIs, in the same format as Table 1. The bottom row is the control. The SAL/jerk crossover observed on Can reproduces here: SAL tracks the length baseline down while jerk rises, and in the band where length is at chance jerk reaches the same 0.896 it reaches in Can’s tightest band.

condition	pool size	best success rate (mean $\pm$ std)	95% CI
oracle (successes only)	716	0.892 $\pm$ 0.048	[0.866, 0.922]
no curation	1950	0.380 $\pm$ 0.063	[0.344, 0.420]
random 50%	1950	0.360 $\pm$ 0.061	[0.326, 0.398]
SAL	1950	0.332 $\pm$ 0.061	[0.298, 0.370]
jerk	1950	0.332 $\pm$ 0.041	[0.306, 0.354]
path length	1950	0.310 $\pm$ 0.030	[0.292, 0.328]
TED	1950	0.240 $\pm$ 0.045	[0.212, 0.266]

Table 3: Downstream GMM-BC policy success on can/mg after filtering to a fixed 1950-demonstration pool by each metric, 10 seeds each. The oracle pool is smaller (716, the number of successes that exist) and is a ceiling reference, not a matched condition. Pairwise tests against the baselines are in Table 4.

203 oracle (716, the number of successes that exist); we therefore treat the oracle as a ceiling reference
204 rather than a matched condition.

205 Table 3 answers the question Section 4 could only pose, and Table 4 reports the comparisons that
206 matter. Two results are unambiguous. Curating by TED costs roughly a third of downstream success
207 relative to random selection of the same size (-0.120 , $p = 0.0001$) and relative to not curating at all
208 (-0.140 , $p < 0.0001$); both survive Bonferroni correction across the eight comparisons we report.
209 Path length is directionally harmful with weaker support ($p = 0.037$ and $p = 0.0071$), clearing
210 conventional significance but not correction. SAL and jerk sit below both baselines in the mean but
211 are not statistically distinguishable from them ($p \geq 0.06$), and we report them as such rather than as
212 evidence of harm.

213 We emphasize this because an earlier version of this analysis, run with 5 seeds rather than 10,
214 appeared to show every metric-curated condition underperforming both baselines. That pattern did
215 not survive the seed expansion for SAL and jerk. Since the argument of this paper is that under-
216 powered evaluation produces confident claims in the wrong direction, we apply the same standard
217 to our own results: the defensible claim is about TED, not about smoothness metrics as a class.

218 What the grid does establish irrespective of any single comparison is that under this policy class none
219 of these metrics recovers usable signal that is demonstrably present. An oracle keeping only true
220 successes reaches 0.892 [0.866, 0.922], roughly $2.5\times$ the best metric-curated condition, so the pool
221 contains ample information about which demonstrations to keep; the trajectory statistics simply
222 do not find it. Random subsampling and the uncurated control are themselves indistinguishable
223 ($\Delta = -0.020$, CI spanning zero), which is the sanity check one wants: at a 50% keep fraction on
224 this data, discarding demonstrations at random costs nothing measurable, so any deficit a metric
225 shows is attributable to *which* demonstrations it discards rather than to the reduction in pool size.

226 The downstream ordering tracks the detection ordering imperfectly. TED is worst in both, and path
227 length is weak in both, but jerk has the best detection AUROC on can/mg (0.706) while landing
228 indistinguishable from SAL downstream, whose detection AUROC is near chance (0.454). Detec-
229 tion accuracy is at best a coarse predictor of curation value, consistent with the decoupling reported
230 by [Bedi, 2026b] on a different testbed.

metric	vs. random 50%		vs. no curation	
	Δ [95% CI]	p	Δ [95% CI]	p
TED	-0.120 [-0.166, -0.076]	0.0001	-0.140 [-0.186, -0.096]	< 0.0001
path length	-0.050 [-0.090, -0.012]	0.037	-0.070 [-0.112, -0.030]	0.0071
jerk	-0.028 [-0.072, +0.014]	0.25	-0.048 [-0.092, -0.006]	0.060
SAL	-0.028 [-0.080, +0.024]	0.32	-0.048 [-0.098, +0.002]	0.098

Table 4: Difference in mean best success rate between each metric-curated condition and the two baselines, with 95% bootstrap CIs on the difference (20,000 resamples) and Welch two-sample p -values. Bold marks the comparisons surviving Bonferroni correction across all eight tests ($\alpha = 0.05/8 = 0.006$). Only TED clears that bar. For reference, random and no-curation are themselves indistinguishable ($\Delta = -0.020$, CI [-0.072, +0.032]).

condition	GMM-BC (10 seeds)	diffusion (5 seeds)
oracle (successes only)	0.892 $\pm$ 0.048	0.544 $\pm$ 0.030
no curation	0.380 $\pm$ 0.063	0.120 $\pm$ 0.014
random 50%	0.360 $\pm$ 0.061	0.116 $\pm$ 0.017
jerk	0.332 $\pm$ 0.041	0.212 $\pm$ 0.030
TED	0.240 $\pm$ 0.045	0.092 $\pm$ 0.030

Table 5: Mean best success rate under both policy classes, restricted to the five conditions run under both. TED is the worst condition under each. Jerk moves from indistinguishable-from-baseline under BC to above both baselines under diffusion. Absolute rates are lower under diffusion in every condition, by factors ranging from $1.6\times$ (oracle) to $3.1\times$ (the two baselines).

5.1 Does the curation decision transfer across policy classes?

The grid above uses one policy class, so it cannot separate a property of the metric from a property of behavior cloning. We therefore replicated it under a diffusion policy on the same curated subsets: same keep fraction, same 1950-demonstration pools, same 600 epochs of 500 gradient steps, same learning rate (10^{-4}), and the same evaluation protocol (50 rollouts, horizon 400, every 150 epochs, best checkpoint per seed).¹ The policy is a conditional U-Net ([256, 512, 1024] down-dims, kernel 5, 8 groups, 256-dimensional diffusion-step embedding) with EMA (power 0.75), an observation horizon of 2, action horizon of 8, and prediction horizon of 16, using a DDIM noise scheduler with 10 inference steps. Batch size (256 vs. 100) and the sequence modeling are inherent to the architecture and are the only respects in which the two grids differ by more than the policy class. We ran five conditions (TED, jerk, and the three references) with five seeds each; SAL and path length were not rerun, so the diffusion comparison covers two of the four metrics.

Three things follow, and they do not all point the same way.

First, TED is the worst condition under both architectures (0.240 under BC, 0.092 under diffusion), which is what one would expect if the deficit is a property of the metric rather than of behavior cloning. But the harm is statistically robust only under BC. Under diffusion the deficit is -0.024 against random and -0.028 against no curation, with intervals that admit zero, and neither survives correction. This is a power statement, not a refutation: at five seeds this grid resolves differences of roughly 0.054, essentially the same threshold we report for the BC grid, and TED’s deficit falls below it. We therefore claim that TED’s *ordering* holds across architectures and that its *harm* is demonstrated under BC only. Descriptively, and not as a tested comparison, the deficit is of broadly similar relative size in the two grids: 37% of the no-curation baseline under BC and 23% under diffusion. We report this only to characterize how the compressed diffusion success scale relates to

¹Reporting the best checkpoint per seed is a maximum over the eight evaluations each run receives, and is therefore biased upward; both grids are evaluated eight times, so the bias is symmetric across architectures. Recomputing every comparison with the *mean* over the eight evaluations instead leaves all conclusions intact, with one exception we record here: TED versus no curation under diffusion reaches $p = 0.0110$ under the mean estimator and would survive this architecture’s correction threshold ($\alpha = 0.0125$), whereas under the reported best-of-eight estimator it does not ($p = 0.11$). TED versus random remains non-significant under both ($p = 0.12$ and $p = 0.17$). We report the conservative version throughout.

metric	vs. random 50%		vs. no curation	
	Δ [95% CI]	p	Δ [95% CI]	p
jerk	+0.096 [+0.068, +0.124]	0.0007	+0.092 [+0.064, +0.116]	0.0011
TED	-0.024 [-0.048, +0.004]	0.17	-0.028 [-0.052, +0.000]	0.11

Table 6: Diffusion-policy grid, same procedure as Table 4: 95% bootstrap CIs on the difference (20,000 resamples) and Welch two-sample p -values. Bold marks comparisons surviving Bonferroni correction across the four tests reported for this architecture ($\alpha = 0.05/4 = 0.0125$). Random and no-curation are again indistinguishable ($\Delta = -0.004$, CI $[-0.020, +0.012]$).

the absolute differences in Table 6; the diffusion interval admits zero, and nothing here is evidence that the two effects are equal.

Second, jerk reverses sign. Under BC it is indistinguishable from both baselines ($p \geq 0.06$); under diffusion it is better than both by +0.096 and +0.092, and both comparisons survive correction. This is the largest effect in the diffusion grid and it points the opposite way from this paper’s headline: at least one crude trajectory statistic, never proposed as a curation metric, has genuine curation value under the right policy class. It is also the second time in this paper that jerk outperforms the metrics motivated against it, the first being detection inside a duration-matched band. We do not have an account of why the architectures differ here, and we report it as an open result rather than a mechanism.

Third, absolute success rates are lower under diffusion in every condition, by factors ranging from $1.6\times$ on the oracle (0.544 vs. 0.892) to $3.1\times$ on the two baselines, at a matched epoch and evaluation budget in which the diffusion runs in fact saw more samples per epoch (batch 256 vs. 100). The gap is therefore not a uniform offset: it is widest exactly on the mixed-quality pools the curation comparison runs on.

One explanation can be ruled out. The diffusion runs show no measurable improvement over the second half of training: averaged across all 25 runs, mean success in the last four evaluations exceeds the first four by +0.017, against a binomial noise floor of 0.044 for 50-rollout evaluations, and only 3 of 25 runs improve by more than one noise standard deviation. Per-run variation across the eight evaluations has a standard deviation $1.01\times$ the noise floor, indistinguishable from sampling noise alone. The identical measurement on the BC grid gives +0.092, with 56 of 70 runs improving and a ratio of $1.46\times$, so the flatness is a property of the diffusion runs and not of our evaluation protocol. Whatever limits these policies is not training duration; they converge early and stop. We record this because under-training is the most natural explanation for the gap and it does not survive contact with the training curves. The consequence for reading Table 5 is concrete: the two columns are not operating at comparable points on the success scale, so cross-architecture comparisons of absolute differences are weaker than within-column comparisons, and the claims above are deliberately about ordering and sign rather than magnitude.

We correct within each architecture, since each grid is a separate family of comparisons. Pooling all twelve tests instead ($\alpha = 0.05/12 = 0.00417$) changes nothing: TED’s two BC comparisons and jerk’s two diffusion comparisons still survive, and path length’s $p = 0.0071$ still does not.

For a post-training pipeline the practical reading is that a curation metric cannot be selected once and reused across adaptation setups. The same filter that is merely inert under one policy class is the best-performing choice under another, and the metric that is most harmful under one is not measurably harmful under the other. Curation and policy class are a joint choice, and reporting a curation result without naming the architecture it was measured on leaves out something that changes the answer.

6 Discussion

Skill and success are different targets. The same metric family gives opposite verdicts depending on the label. Under an uncontrolled skill label, SAL looks nearly perfect; under a length-controlled skill label it is marginal; under a success label on fixed-length data it is at chance and TED is inverted. Curation papers validate against operator skill or smoothness-adjacent labels on success-only pools,

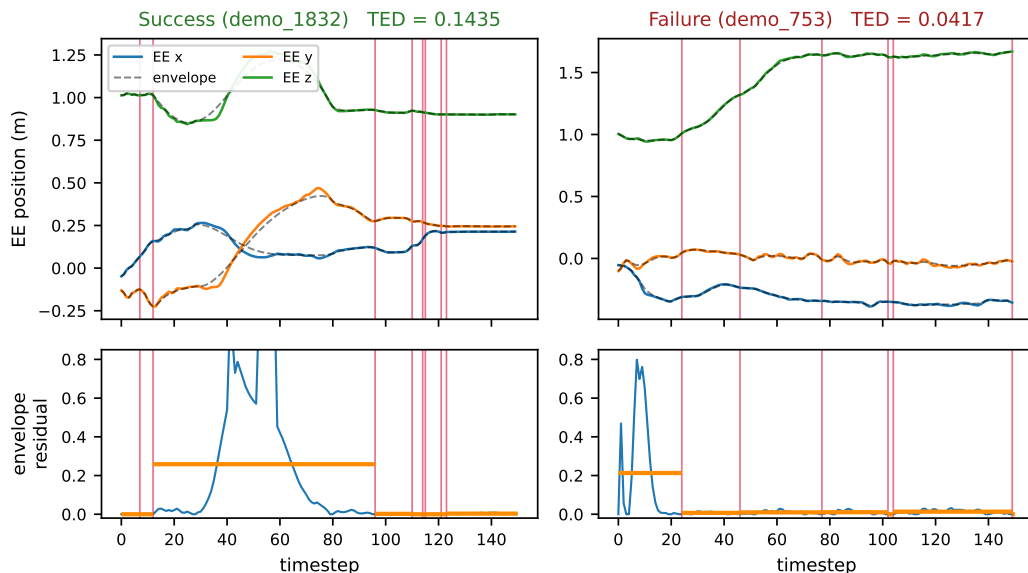


Figure 2: **Failure is smooth; success is not.** A successful (left) and failed (right) can/mg demonstration, with end-effector position, TED’s gripper-contact segment boundaries (red), the per-segment Bézier envelope (dashed), and the per-timestep envelope residual (bottom, orange bars are per-segment means). Both trajectories contain one high-deviation segment, but in the success it spans timesteps 12–96 and coincides with the descend-and-grasp, while in the failure it is confined to the initial free-space approach (0–24) and the remaining 126 steps are a flat tail of repeated gripper cycles that never completes a placement. TED averages over the alignment path and so rates the failure as the cleaner trajectory. Illustrative of the mechanism only; the quantitative claim is Figure 1b.

then are deployed, in practice, to remove *failures* from mixed pools. Our results say those are not the same job, and the metrics do not transfer between them.

The crude baseline wins under controls. RINSE motivates SAL and TED as principled improvements over naïve statistics such as jerk. Under every length control we applied, that ordering reverses. We do not claim jerk is a good curation metric in general, and the Bedi testbeds [Bedi, 2026a] show smoothness-family scorers failing badly on structural defects; we claim only that on real robomimic data the published sophistication does not survive the controls that the audit literature has shown to be necessary.

Why TED inverts: failure is smooth. TED’s design premise is that partitioning at gripper-contact boundaries isolates task-relevant abruptness, and a plausible hypothesis going in was that this would let TED catch failures, since canonical structural failures are contact-boundary events. The data supports a different and simpler mechanism. Figure 2 contrasts a successful and a failed can/mg demonstration. Both contain one high-deviation segment, but they differ in when it occurs and how much of the episode it spans. In the success, the high-residual segment runs from timestep 12 to 96 (84 of 150 steps, mean residual 0.259) and coincides with the descend-and-grasp: the end effector dips, the lateral axes swing, and the envelope cannot track the motion. In the failure, the only comparable deviation is the initial free-space approach (timesteps 0–24, mean residual 0.213); after that the trajectory settles into a flat tail in which the end effector holds height while the gripper opens and closes repeatedly without completing a placement, and every remaining segment has residual below 0.014. Because TED averages deviation over the DTW alignment path, a committed motion spanning half the episode raises the score substantially while a short approach transient is diluted by the smooth remainder. TED therefore rates the failure as the cleaner trajectory (0.042 vs. 0.144).

On this task, in other words, successful manipulation *requires* sustained high-deviation motion through the contact phase, while failure manifests as smooth hovering rather than as erratic thrashing. A metric that rewards smoothness is measuring commitment with the sign flipped. This also explains why raw jerk is the one metric that retains signal in Section 4: jerk is high precisely in the

trajectories that commit. We stress that these panels illustrate a mechanism rather than establish it; the quantitative claim is the AUROC of 0.376 in Figure 1b, and the per-demonstration TED scores we inspected (successes 0.144 and 0.079, failures 0.042, 0.042, and 0.017) are consistent with but not evidence for that number.

7 Limitations

Our detection results cover two robomimic tasks (Can and Square) but our success-label and downstream-training results cover only Can, because Can is the only robomimic task shipping failure-containing splits; extending the success-label analysis requires moving to a different benchmark such as LIBERO. Our downstream results cover two policy classes (GMM behavior cloning and a diffusion policy); the detection results are policy-free. The cross-architecture grid of Section 5.1 carries three limitations of its own. It covers two of the four metrics, since SAL and path length were not rerun under diffusion, so our claim that TED ranks worst under both architectures is made over a five-condition subset on the diffusion side and a seven-condition grid on the BC side. It uses five seeds rather than ten, which resolves differences of roughly 0.054; TED’s diffusion deficit (-0.028 , CI $[-0.052, +0.000]$) falls below that, so we report TED’s ordering as replicating and its harm as established under BC only. And absolute success rates remain lower under diffusion in every condition at a matched epoch and evaluation budget, from $1.6\times$ on the oracle (0.544 vs. 0.892) to $3.1\times$ on the two baselines. An earlier reduced-compute pilot showed the same gap and we attributed it to under-convergence; restoring the full epoch and rollout budget did not close it, and the training curves rule that explanation out directly, since the diffusion runs improve by only $+0.017$ between the first and second halves of training against a 0.044 noise floor while the BC runs improve by $+0.092$. The gap is not a matter of training duration, and we have no replacement explanation for it. Because the two grids sit at different points on the success scale, we restrict cross-architecture claims to ordering and sign rather than magnitude. Multiple-comparison correction is applied within each architecture, as each grid is a separate family; pooling all twelve tests ($\alpha = 0.00417$) leaves every reported conclusion unchanged. Our TED is a reimplementation from the RINSE appendix, not the authors’ released code; as a sanity check it broadly agrees with SAL’s ranking on the clean proficient-human data, but a discrepancy with the reference implementation cannot be excluded, and we will replace it if code becomes available. The tightest duration band retains only 7 negatives, and its role in our argument is corroborative, not primary. The BC downstream conditions use 10 seeds; at this power, differences smaller than roughly 0.05 in mean success rate are not resolvable, which is why we report SAL and jerk as indistinguishable from the baselines rather than as harmful under that architecture. The oracle training condition has a smaller pool (716 vs. 1950) and is reported as a ceiling, not a matched comparison. Finally, our success-label experiments use machine-generated failures (can/mg); failure modes of human teleoperators may present differently, and extending the audit to human-failure pools is the clearest next step. We report can/paired only as a secondary setting because its matched-pair construction, in which each success/failure pair shares a near-identical trajectory prefix and diverges only in the final placement, concentrates the discriminating signal into a handful of terminal timesteps and simultaneously introduces an opposing length confound (successful demonstrations are systematically longer, length-alone AUROC 0.194); neither property is representative of the mixed-quality pools curation metrics are deployed on.

8 Conclusion

Published trajectory-smoothness curation metrics, evaluated exactly where their authors did not, on real operator labels under duration controls and on fixed-length data against task success, rank episode duration more than demonstration quality. Once duration is controlled, the crude jerk baseline they were designed to supersede outperforms them, and the contact-aware metric inverts, and the same crossover reproduces on a second task without retuning. Training on the metrics’ own curated subsets shows the practical stakes are real but narrower than a five-seed pass suggested: curating by TED costs roughly a third of downstream success against random selection of the same size, a result robust to correction for multiple comparisons, while the other metrics are directionally below the baselines without being distinguishable from them. An oracle reaches 0.892 on the same pool, so the signal these metrics fail to find is unambiguously present in the data. Replicating the grid under a diffusion policy leaves TED ranked worst under both architectures, though at five seeds its deficit there is not resolvable, and reverses jerk from inert to better than both baselines. For a

375 post-training pipeline this is the operative point: which demonstrations are worth keeping is not a
376 property of the demonstrations alone, and a curation metric carried unchanged across policy classes
377 can change sign.

378 **Reproducibility.** All datasets are the public robomimic release [Mandlekar et al., 2021]; SAL
379 follows its published formula and TED is reimplemented from the RINSE appendix as described in
380 Section 3. Metric implementations, scoring scripts, and training configurations will be released at an
381 anonymized repository upon acceptance and de-anonymized in the camera-ready or public posting.

382 **Use of AI assistance.** An AI assistant was used for code scaffolding, LaTeX drafting, and editing
383 support during this project. All experiments were designed, executed, and verified by the author(s),
384 and all reported numbers were produced by the released code and checked against the raw outputs.

385 References

- 386 S. Kulkarni, R. Dhar, Y. Cui. Learning from the Best: Smoothness-Driven Metrics for Data Quality
387 in Imitation Learning. arXiv:2604.23000, 2026.
- 388 A. Bedi. Auditing Demonstration Curation Metrics: Action-Only Scorers Fail on the Structural
389 Defects That Degrade Imitation Policies. arXiv:2606.05588, 2026.
- 390 A. Bedi. What Demonstration Curation Metrics Do to Your Policy. arXiv:2606.10229, 2026.
- 391 A. Bedi. Phase-Localized Curation Does Not Help: A Negative Result on Per-Phase Metric Selec-
392 tion for Demonstration Filtering. arXiv:2606.15064, 2026.
- 393 N. Sojib, M. Begum. An Efficient Metric for Data Quality Measurement in Imitation Learning.
394 arXiv:2605.01544, 2026.
- 395 A. Mandlekar et al. What Matters in Learning from Offline Human Demonstrations for Robot
396 Manipulation. CoRL, 2021.
- 397 H. Kress-Gazit et al. Robot Learning as an Empirical Science: Best Practices for Policy Evaluation.
398 arXiv:2409.09491, 2024.
- 399 Y. R. Wang, C. Ung, C. Tan, G. Tannert, J. Duan, J. Li, A. Le, R. Oswal, M. Grotz, W. Pumacay,
400 Y. Deng, R. Krishna, D. Fox, S. Srinivasa. RoboEval: Where Robotic Manipulation Meets Struc-
401 tured and Scalable Evaluation. arXiv:2507.00435, 2026.
- 402 T. Jiang, X. Tan, S. Wheeler, L. Sun, T. W. Ayalew, M. Walter. What Are We Actually Benchmarking
403 in Robot Manipulation? arXiv:2606.04233, 2026.